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CS3451 - Introduction to Operating Systems

Topic: scheduling

1. SUMMARY

Scheduling is the process of allocating system resources to computer programs or processes. The operating system acts as a resource allocator, deciding how to allocate resources such as CPU time, memory space, storage space, and I/O devices to specific programs and users. The operating system provides the means for proper use of these resources in the operation of the computer system. There are different types of scheduling policies and algorithms, such as First-Come-First-Served (FCFS), Shortest Job First (SJF), Priority Scheduling, Round Robin (RR), and Multi-Level Feedback Queue (MLFQ). Scheduling is essential for efficient and fair resource allocation. The operating system's view of scheduling is as a resource allocator, whereas the user's view is as a performance and security enhancer.

2. SHORT NOTES

Detailed Definition with Examples

Scheduling is a process by which the operating system allocates system resources to computer programs or processes. This allocation is done to ensure efficient and fair use of the system resources.

Types and Classifications

1. **FCFS (First-Come-First-Served) Scheduling**: In FCFS, the process that arrives first is executed first. This is the simplest scheduling algorithm and is suitable for systems where processes arrive at regular intervals.
2. **SJF (Shortest Job First) Scheduling**: In SJF, the process with the shortest burst time is executed first. This algorithm is suitable for systems where the length of the process is unpredictable.

3. **Priority Scheduling**: In priority scheduling, processes are assigned a priority level based on their importance. The process with the highest priority is executed first.
4. **RR (Round Robin) Scheduling**: In RR, each process is assigned a fixed time slice (called a time quantum) to execute. The process that completes its execution within the time quantum is given the next time quantum, and the process that does not complete its execution within the time quantum is moved to the end of the queue.
5. **MLFQ (Multi-Level Feedback Queue) Scheduling**: In MLFQ, multiple queues with different time quanta are used to schedule processes.

Working Principle

The working principle of scheduling involves the following steps:

1. **Process Arrival**: A process arrives at the operating system.
2. **Process Selection**: The operating system selects a process to execute based on the chosen scheduling algorithm.
3. **Process Execution**: The selected process is executed for a certain time period.
4. **Process Termination**: The process completes its execution and terminates.

Important Formulas and Equations

1. **Average Waiting Time (AWT)**: AWT = (? (BTi * Ti)) / (? Ti)
2. **Average Turnaround Time (ATT)**: ATT = (? (BTi + Ti)) / (? Ti)
3. **Throughput**: Throughput = (Number of processes completed) / (Total time taken)

Diagrams Description

The diagram of a computer system consists of the following components:

1. **CPU**: Central Processing Unit
2. **Memory**: Main Memory

3. **I/O Devices**: Input/Output Devices

4. **Operating System**: The operating system acts as a resource allocator and schedules processes to execute.

Advantages and Disadvantages

Advantages:

1. **Efficient Resource Allocation**: Scheduling ensures efficient allocation of system resources.

2. **Fair Resource Allocation**: Scheduling ensures fair allocation of system resources.

3. **Improved System Performance**: Scheduling improves system performance by allocating resources to processes that need them the most.

Disadvantages:

1. **Complexity**: Scheduling algorithms can be complex to implement.

2. **Overhead**: Scheduling algorithms can incur overhead due to context switching and process switching.

Real-World Applications

Scheduling is used in various real-world applications, such as:

1. **Operating Systems**: Scheduling is used in operating systems to allocate system resources to processes.

2. **Networks**: Scheduling is used in networks to allocate network resources to processes.

3. **Databases**: Scheduling is used in databases to allocate database resources to queries.

Comparison with Related Concepts

Scheduling can be compared with other related concepts, such as:

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1. **Multiprogramming**: Multiprogramming involves running multiple programs concurrently.

2. **Multiprocessing**: Multiprocessing involves running multiple programs on multiple processors concurrently.

3. **Concurrency**: Concurrency involves executing multiple processes or threads concurrently.

Time and Space Complexity

The time complexity of scheduling algorithms varies depending on the algorithm used. Some scheduling algorithms have a time complexity of O(n), while others have a time complexity of O(n log n) or even O(n^2).

3. PRACTICAL / CODING EXAMPLES

Since this is a theoretical subject, no hands-on implementation is provided.

4. IMPORTANT QUESTIONS

Part-A (2 Marks) ? 5 questions, each with a crisp 2-3 line answer:

Q1. Define scheduling and list its types.

Answer: Scheduling refers to the process of allocating system resources to tasks or processes. The types of scheduling include Preemptive, Non-Preemptive, First-Come-First-Served (FCFS), Shortest Job First (SJF), Priority Scheduling, Round Robin (RR), and Multilevel Queue Scheduling.

Q2. Compare preemptive and non-preemptive scheduling.

Answer: Preemptive scheduling allows the operating system to interrupt a currently running process, whereas non-preemptive scheduling does not. Preemptive scheduling is more efficient in terms of response time, but it is more complex to implement.

Q3. What is the purpose of scheduling in an operating system?

Answer: The primary purpose of scheduling is to manage system resources efficiently, maximize throughput, and minimize response time. It ensures that tasks are executed in a fair and orderly manner, preventing conflicts and deadlocks.

Q4. Write a short note on First-Come-First-Served (FCFS) scheduling.

Answer: FCFS is a non-preemptive scheduling algorithm that executes tasks in the order they arrive in the ready queue. It is simple to implement but can lead to poor performance if a long-running task is executed first, causing other tasks to wait.

Q5. Give an example of scheduling in a real-world application.

Answer: Scheduling is used in operating systems to manage processes, in embedded systems to control tasks, and in web servers to handle requests. For instance, a web server uses scheduling to allocate CPU time to different requests, ensuring that each request is processed efficiently and fairly.

Part-B (13 Marks) ? 3 questions. Each answer MUST be at least 20-25 lines long.

Q1. Explain scheduling in detail with neat diagram and suitable examples.

Answer:

Scheduling is the process of allocating system resources to tasks or processes. It is a crucial component of operating systems, as it ensures that tasks are executed efficiently and fairly. The primary goal of scheduling is to maximize throughput, minimize response time, and prevent conflicts and deadlocks.

There are several types of scheduling, including:

1. Preemptive Scheduling: The operating system can interrupt a currently running process.
2. Non-Preemptive Scheduling: The operating system cannot interrupt a currently running process.
3. First-Come-First-Served (FCFS) Scheduling: Tasks are executed in the order they arrive in the ready queue.
4. Shortest Job First (SJF) Scheduling: The task with the shortest execution time is executed first.
5. Priority Scheduling: Tasks are assigned priorities, and the task with the highest priority is executed first.

The scheduling process involves the following steps:

1. Process creation: A new process is created, and its attributes, such as priority and execution time, are set.
2. Process scheduling: The scheduler selects the next process to execute based on the scheduling algorithm.
3. Process execution: The selected process is executed, and its state is updated.
4. Process termination: The process is terminated, and its resources are released.

The following diagram illustrates the scheduling process:

```
  +---------------+
  |  Process      |
  |  Creation     |
  +---------------+
          |
          |
          v
  +---------------+
  |  Process      |
  |  Scheduling   |
  +---------------+
          |
          |
          v
  +---------------+
  |  Process      |
  |  Execution    |
  +---------------+
          |
          |
          v
  +---------------+
  |  Process      |
```

| Termination|

+---------------+

```

Scheduling has several advantages, including:

* Improved system performance

* Increased throughput

* Reduced response time

* Fairness and predictability

However, scheduling also has some disadvantages, including:

* Complexity: Scheduling algorithms can be complex to implement and manage.

* Overhead: Scheduling can introduce overhead, such as context switching and process creation.

Scheduling is used in various real-world applications, including:

* Operating systems: Scheduling is used to manage processes and threads.

* Embedded systems: Scheduling is used to control tasks and manage resources.

* Web servers: Scheduling is used to handle requests and allocate CPU time.

In conclusion, scheduling is a critical component of operating systems, and its primary goal is to allocate system resources efficiently and fairly. There are several types of scheduling, each with its advantages and disadvantages. Scheduling is used in various real-world applications, and its importance cannot be overstated.

Q2. Describe the types of scheduling with working principle and examples.

Answer:

There are several types of scheduling, each with its working principle and examples. The following are some of the most common types of scheduling:

1. First-Come-First-Served (FCFS) Scheduling: This type of scheduling executes tasks in the order they arrive in the ready queue. The working principle of FCFS is simple: the task that arrives first is executed first.

Example: A print queue uses FCFS scheduling to print documents in the order they are received.

2. Shortest Job First (SJF) Scheduling: This type of scheduling executes the task with the shortest execution
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time first. The working principle of SJF is to minimize the average waiting time of tasks.

Example: A web server uses SJF scheduling to handle requests, executing the request with the shortest execution time first.

3. Priority Scheduling: This type of scheduling assigns priorities to tasks and executes the task with the highest priority first. The working principle of priority scheduling is to ensure that critical tasks are executed promptly.

Example: An operating system uses priority scheduling to execute system processes, assigning higher priorities to critical processes.

4. Round Robin (RR) Scheduling: This type of scheduling executes tasks in a circular fashion, allocating a fixed time slice to each task. The working principle of RR is to ensure that each task receives a fair share of CPU time.

Example: A time-sharing system uses RR scheduling to allocate CPU time to multiple users.

The following comparison table summarizes the types of scheduling:

Type	Working Principle	Example
FCFS	Execute tasks in the order they arrive	Print queue
SJF	Execute the task with the shortest execution time	Web server
Priority	Execute the task with the highest priority	Operating system
RR	Execute tasks in a circular fashion	Time-sharing system

Each type of scheduling has its advantages and disadvantages. For example, FCFS is simple to implement but can lead to poor performance if a long-running task is executed first. SJF can minimize the average waiting time, but it requires knowledge of task execution times. Priority scheduling can ensure that critical tasks are executed promptly, but it can be complex to manage priorities.

In conclusion, the types of scheduling are designed to allocate system resources efficiently and fairly. Each type has its working principle and examples, and the choice of scheduling algorithm depends on the specific application and requirements.

Q3. Compare scheduling with related concepts. Discuss advantages and applications.

Answer:

Scheduling is related to several concepts, including:

1. Process management: Scheduling is a critical component of process management, as it ensures that processes are executed efficiently and fairly.

2. Resource allocation: Scheduling is related to resource allocation, as it involves allocating system resources to tasks and processes.

3. Deadlock prevention: Scheduling is related to deadlock prevention, as it can prevent deadlocks by ensuring that tasks are executed in a fair and orderly manner.

The following comparison table summarizes the related concepts:

Concept	Description	Advantages	Applications
Process management	Manage processes and threads	Improved system performance, increased throughput	Operating systems, embedded systems
Resource allocation	Allocate system resources to tasks	Efficient use of resources, improved system performance	Operating systems, web servers
Deadlock prevention	Prevent deadlocks by ensuring fair and orderly execution	Prevents system crashes, ensures predictability	Operating systems, database systems

Scheduling has several advantages, including:

* Improved system performance
* Increased throughput
* Reduced response time
* Fairness and predictability

Scheduling is used in various applications, including:

* Operating systems: Scheduling is used to manage processes and threads.
* Embedded systems: Scheduling is used to control tasks and manage resources.
* Web servers: Scheduling is used to handle requests and allocate CPU time.

In conclusion, scheduling is a critical component of operating systems, and its related concepts include process management, resource allocation, and deadlock prevention. Scheduling has several advantages and is used in various applications, including operating systems, embedded systems, and web servers.

Part-C (15 Marks) ? 2 questions. Each answer MUST be at least 30 lines long.

Q1. Elaborate on scheduling covering theory, implementation, real-world applications, and future scope.

Answer:

Section 1: Deep theoretical background

Scheduling is a fundamental concept in operating systems, and its theory is based on the idea of allocating system resources to tasks and processes. The primary goal of scheduling is to maximize throughput, minimize response time, and prevent conflicts and deadlocks. There are several scheduling algorithms, including First-Come-First-Served (FCFS), Shortest Job First (SJF), Priority Scheduling, and Round Robin (RR).

The theory of scheduling involves the study of scheduling algorithms, their properties, and their applications. Scheduling algorithms can be classified into two categories: preemptive and non-preemptive. Preemptive scheduling allows the operating system to interrupt a currently running process, whereas non-preemptive scheduling does not.

Section 2: How it is implemented/used in practice

Scheduling is implemented in operating systems using a combination of hardware and software components. The scheduler is responsible for selecting the next process to execute, and it uses various algorithms to make this decision. The scheduler also manages the process table, which contains information about each process, such as its state, priority, and execution time.

In practice, scheduling is used in various applications, including operating systems, embedded systems, and web servers. For example, an operating system uses scheduling to manage processes and threads, allocating CPU time and other resources to each process. An embedded system uses scheduling to control tasks and manage resources, ensuring that the system operates efficiently and reliably.

Section 3: Real-world case studies or applications

There are several real-world case studies or applications of scheduling, including:

1. Google's Borg system: Borg is a cluster management system that uses scheduling to allocate resources to tasks and applications. Borg uses a combination of scheduling algorithms, including FCFS and priority scheduling, to ensure that tasks are executed efficiently and fairly.
2. Apache HTTP Server: The Apache HTTP Server uses scheduling to handle requests and allocate CPU time to each request. The server uses a combination of scheduling algorithms, including RR and priority scheduling, to ensure that requests are handled efficiently and fairly.
3. Linux operating system: The Linux operating system uses scheduling to manage processes and threads, allocating CPU time and other resources to each process. Linux uses a combination of scheduling algorithms, including FCFS, SJF, and priority scheduling, to ensure that processes are executed efficiently and fairly.

Section 4: Limitations and challenges

Scheduling has several limitations and challenges, including:

1. Complexity: Scheduling algorithms can be complex to implement and manage, especially in large-scale systems.
2. Overhead: Scheduling can introduce overhead, such as context switching and process creation, which can affect system performance.
3. Fairness: Scheduling algorithms must ensure fairness and predictability, which can be challenging in systems with varying workloads and priorities.

Section 5: Future scope and improvements

The future scope of scheduling includes the development of new scheduling algorithms and techniques, such as:

1. Machine learning-based scheduling: Machine learning algorithms can be used to predict task execution times and optimize scheduling decisions.
2. Real-time scheduling: Real-time scheduling algorithms can be used to ensure that tasks are executed within strict deadlines and constraints.
3. Distributed scheduling: Distributed scheduling algorithms can be used to allocate resources to tasks and applications in distributed systems.

In conclusion, scheduling is a critical component of operating systems, and its theory, implementation, and applications are diverse and complex. Scheduling has several limitations and challenges, but its future scope includes the development of new scheduling algorithms and techniques that can improve system performance, efficiency, and reliability.

Q2. Critically analyze scheduling. Compare approaches, evaluate trade-offs, and justify with examples.

Answer:

Section 1: Overview of the concept and its importance

Scheduling is a fundamental concept in operating systems, and its importance cannot be overstated. Scheduling is responsible for allocating system resources to tasks and processes, ensuring that they are executed efficiently and fairly. The primary goal of scheduling is to maximize throughput, minimize response time, and prevent conflicts and deadlocks.

Section 2: Different approaches/techniques used

There are several approaches and techniques used in scheduling, including:

1. First-Come-First-Served (FCFS) Scheduling: FCFS is a simple and intuitive scheduling algorithm that executes tasks in the order they arrive in the ready queue.
2. Shortest Job First (SJF) Scheduling: SJF is a scheduling algorithm that executes the task with the shortest execution time first.
3. Priority Scheduling: Priority scheduling is a scheduling algorithm that assigns priorities to tasks and executes the task with the highest priority first.
4. Round Robin (RR) Scheduling: RR is a scheduling algorithm that executes tasks in a circular fashion, allocating a fixed time slice to each task.

Section 3: Detailed comparison of approaches with a table

The following table compares the different scheduling approaches:

Approach	Description	Advantages	Disadvantages
FCFS	Execute tasks in the order they arrive	Simple to implement, fair	Poor performance if long-running task is executed first
SJF	Execute the task with the shortest execution time	Minimizes average waiting time	Requires

knowledge of task execution times |

| Priority | Execute the task with the highest priority | Ensures critical tasks are executed promptly | Can be complex to manage priorities |

| RR | Execute tasks in a circular fashion | Ensures each task receives a fair share of CPU time | Can be complex to implement and manage |

**Section 4: Trade-off analysis (time, space, cost, complexity

5. MCQs

Q1. What is the primary function of the operating system in a computer system?

a) To execute user programs

b) To control hardware resources

c) To provide a user interface

d) To manage system resources

Answer: b) To control hardware resources

Explanation: The operating system acts as a resource allocator and controls hardware resources.

Q2. Which scheduling algorithm is suitable for systems where processes arrive at regular intervals?

a) FCFS (First-Come-First-Served)

b) SJF (Shortest Job First)

c) Priority Scheduling

d) RR (Round Robin)

Answer: a) FCFS (First-Come-First-Served)

Explanation: FCFS is suitable for systems where processes arrive at regular intervals.

Q3. What is the time complexity of the RR (Round Robin) scheduling algorithm?

a) O(n)

b) O(n log n)

c) O(n^2)

d) Variable

Answer: a) O(n)

Explanation: The time complexity of RR is O(n), where n is the number of processes.

Q4. What is the primary advantage of the MLFQ (Multi-Level Feedback Queue) scheduling algorithm?

a) Efficient resource allocation

b) Fair resource allocation

c) Improved system performance

d) Reduced overhead

Answer: d) Reduced overhead

Explanation: MLFQ reduces overhead by using multiple queues with different time quanta.

Q5. Which scheduling algorithm is suitable for systems where the length of the process is unpredictable?

a) FCFS (First-Come-First-Served)

b) SJF (Shortest Job First)

c) Priority Scheduling

d) RR (Round Robin)

Answer: b) SJF (Shortest Job First)

Explanation: SJF is suitable for systems where the length of the process is unpredictable.

7. YOUTUBE LINKS

- [Scheduling Tutorial](https://www.youtube.com/results?search_query=scheduling+tutorial)
- [FCFS Scheduling Tutorial](https://www.youtube.com/results?search_query=fcfs+scheduling+tutorial)
- [SJF Scheduling Tutorial](https://www.youtube.com/results?search_query=sjf+scheduling+tutorial)
- [Priority Scheduling Tutorial](https://www.youtube.com/results?search_query=priority+scheduling+tutorial)
- [RR Scheduling Tutorial](https://www.youtube.com/results?search_query=rr+scheduling+tutorial)